

Rational box smoothing with polynomial-denominator conditional means

Proof independently reviewed by `size_bounds` and `independent_directions`

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Throughout, “polynomial” means bounded by a fixed power of $m+1$, with constants allowed to depend on fixed parameters but not on m . The statement keeps the number of auxiliary rank features fixed. It does not bound denominators of higher conditional moments.

Theorem 1. *Fix h . Suppose K is polynomial in $m \geq 2$, the nodes $0 < b_1 < \dots < b_K < 1$ have gaps, including endpoint gaps, at least an inverse polynomial, and $\psi_{\ell q}$, $1 \leq \ell \leq h$, are rational features with polynomially bounded absolute values and polynomial common denominator. Suppose*

$$T_{\ell j} = K^{-1} \sum_{q=1}^K \psi_{\ell q} b_q^j \quad (0 \leq j \leq m)$$

are rational, and that the h numbers $T_{\ell 1}$ have polynomial common denominator. Finally, let ϕ_j be the orthonormal shifted Legendre polynomials and assume the evaluation family

$$\psi_{\ell q} \phi_j(b_q), \quad 1 \leq \ell \leq h, \quad 0 \leq j < m,$$

has empirical Gram matrix bounded below by an inverse polynomial.

Then there exist disjoint rational intervals $I_q = [c_q - \epsilon, c_q + \epsilon] \subset (0, 1)$, with positive ϵ an inverse polynomial and with all their endpoints on one polynomial rational grid, and rational numbers a_q with $|a_q| \leq 1/2$, such that the probability measure assigning each I_q mass $1/K$ and conditional density

$$\frac{1}{2\epsilon} \left[1 + a_q \left(\frac{(x - c_q)^2}{\epsilon^2} - \frac{1}{3} \right) \right] \tag{1}$$

satisfies

$$\sum_q \psi_{\ell q} \int_{I_q} x^j d\beta(x) = T_{\ell j} \quad (0 \leq j \leq m).$$

Each box has conditional mean exactly c_q , hence these means have a polynomial common denominator. All conditional moments are rational. The centers can be made closer than any prescribed inverse polynomial to the original nodes by adjusting the fixed exponents.

Proof. All spaces below have polynomial dimension. Polynomial evaluation, first- and higher fixed-order derivative norms on $[0, 1]$ are polynomial in m ; intervals will remain inside $[0, 1]$. We first replace the atoms by *uniform* boxes and solve the resulting moment problem over the reals, before rational rounding.

Choose rational $\epsilon = (m+1)^{-A}$ with A a sufficiently large fixed integer. For centers c , consider the nonconstant moment map of the uniform boxes with feature weights $\psi_{\ell q}$. Use the basis $f_j(t) = \int_0^t \phi_{j-1}(u) du$, $1 \leq j \leq m$, instead of powers. Its derivative at zero box width and centers b is K^{-1} times the stated evaluation family. The Gram hypothesis gives a fixed right inverse with polynomial norm from Euclidean residuals to maximum center displacement. Uniform-box averaging changes each baseline moment by $O(\text{poly}(m)\epsilon^2)$, since its first centered moment

is zero. Its Jacobian differs by the same type of bound. The usual fixed-inverse contraction therefore gives real centers

$$c_q^* = b_q + O(\text{poly}(m)\epsilon^2)$$

whose uniform boxes satisfy every target moment exactly. Choose A large enough to preserve all node and endpoint gaps. The derivative bounds in this contraction are polynomial; no denominator assertion is made about c^* .

Rounding while preserving all first moments. The degree-one equations are exactly

$$K^{-1} \sum_q \psi_{\ell q} c_q = T_{\ell 1}. \quad (2)$$

The feature matrix $(\psi_{\ell q})$ has rank h and polynomial Gram lower bound, as a subfamily of the assumed evaluation matrix. Since h is fixed, Cauchy–Binet supplies an $h \times h$ column minor with polynomially bounded inverse. Round every c_q^* to a multiple of $1/L$, where $L = (m+1)^B$ with B a sufficiently large fixed integer, and correct only these h selected centers to restore (2). The resulting centers c_q are rational and

$$\max_q |c_q - c_q^*| \leq \text{poly}(m)/L.$$

They have a polynomial common denominator: the feature entries and targets in (2) have polynomial common denominators, the minor has fixed size, and Cramer’s rule multiplies only a fixed number of such denominators and bounded determinant numerators. Increasing B retains disjointness of the rational boxes and all endpoint gaps.

Correcting higher moments without moving the means. Use for the remaining equations the rationally rescalable basis

$$g_j(t) = \int_0^t (t-u) \phi_{j-2}(u) du, \quad 2 \leq j \leq m.$$

Together with $1, t$, these span $\mathcal{P}_m$. Add the density corrections in (1); these preserve each box’s mass and mean, because for uniform U on $[-1, 1]$,

$$\mathbb{E}(U^2 - 1/3) = \mathbb{E}[U(U^2 - 1/3)] = 0, \quad \mathbb{E}[U^2(U^2 - 1/3)] = 4/45.$$

The response matrix V of the remaining equations is exactly linear in the a_q . Taylor expansion gives

$$V_{(\ell, j), q} = \frac{\psi_{\ell q}}{K} \mathbb{E}[g_j(c_q + \epsilon U)(U^2 - 1/3)] = \frac{2\epsilon^2}{45K} \psi_{\ell q} \phi_{j-2}(c_q) + O(\text{poly}(m)\epsilon^4/K).$$

The Gram hypothesis, center closeness, and sufficiently large A therefore give a right inverse of V with norm at most $\epsilon^{-2} \text{poly}(m)$. The moment residual after center rounding is only $\text{poly}(m)/L$, because the uniform boxes at c^* were exact. Consequently the linear system has a solution with

$$\|a\|_\infty \leq \text{poly}(m)/(L\epsilon^2) \leq 1/2$$

if B is sufficiently large after A is fixed.

The solution may be taken rational without sacrificing this bound. Indeed, rescale each g_j to remove its normalizing square root; then V and the residual vector are rational. The minimum Euclidean norm solution $V^\top(VV^\top)^{-1}e$ is rational. It is unchanged by any invertible rescaling of the equation rows, so the polynomial bounds established in the orthonormal basis still apply.

The bracket in (1) lies between $2/3$ and $4/3$, proving positivity. It is even about c_q , so its conditional mean is c_q . Rational coefficients and endpoints give rational conditional moments. All required equations hold because the constant and linear ones were preserved separately and the remaining spanning equations were solved. Finally, increasing A and then B gives any fixed inverse-polynomial center accuracy while all denominators claimed in the theorem remain polynomial. $\square$

Theorem 2 (A fixed number of polynomial-denominator moments). *Fix integers $h, s \geq 1$, and assume $m \geq s+1$. Under the hypotheses of Theorem 1, strengthen the arithmetic assumption by requiring all $T_{\ell j}$, $1 \leq \ell \leq h$, $1 \leq j \leq s$, to have a polynomial common denominator. Then there are rational boxes on one polynomial grid and positive rational conditional densities of degree at most $s+1$, preserving all the target weighted moments, for which all conditional box moments through degree s have one polynomial common denominator. Each box has mass $1/K$, and the center accuracy can be prescribed as before. The conditional mean need not equal the box center in this extension. Density bounds may be taken to be $1/2$ and $3/2$ relative to the uniform density of the box.*

Proof. First solve the real uniform-box problem exactly, obtaining the centers c_q^* as before. Let μ_{qj}^* be the conditional moment of degree j of the uniform box centered at c_q^* . Choose rational centers c_q on a fine common polynomial grid, and keep the common rational halfwidth ϵ . For each $1 \leq j \leq s$, round all the numbers μ_{qj}^* to a common polynomial grid and correct the same fixed h selected entries to impose

$$K^{-1} \sum_q \psi_{\ell q} \mu_{qj} = T_{\ell j}, \quad 1 \leq \ell \leq h.$$

The fixed-size minor argument proves that all μ_{qj} have a polynomial common denominator and differ from μ_{qj}^* by at most $\text{poly}(m)/L$. There are only the fixed number hs of additional corrected entries. Take $\mu_{q0} = 1$.

In the uniform coordinate $u = (x - c_q)/\epsilon$, use the first s Legendre modes to realize these conditional moments:

$$1 + \sum_{l=1}^s a_{ql} P_l(u).$$

For raw moment j , the response of mode l is $\mathbb{E}[(c_q + \epsilon U)^j P_l(U)]$. It vanishes when $j < l$, and at $j = l$ equals $\epsilon^l \mathbb{E}[U^l P_l(U)]$, a nonzero rational constant times ϵ^l . This fixed-size triangular system has rational inverse of norm at most $C_s \epsilon^{-s}$ for $0 \leq c_q \leq 1$. Thus it realizes the prescribed moments with

$$|a_{ql}| \leq \text{poly}(m)/(L\epsilon^s).$$

They may be made as small as desired by increasing the polynomial exponent in L .

It remains to restore all higher global moments. These low-mode adjustments and center rounding have produced residual at most $\text{poly}(m)/(L\epsilon^s)$ in a polynomially bounded basis. Add one further coefficient $a_{q,s+1} P_{s+1}(u)$ on each box. It preserves every already fixed conditional moment of degree at most s . For the remaining global equations use functions g_j , $j = s+1, \dots, m$, whose $(s+1)$ st derivatives are respectively $\phi_0, \dots, \phi_{m-s-1}$ and whose first $s+1$ Taylor coefficients at zero vanish. The response matrix divided by ϵ^{s+1} converges, with polynomial error $O(\text{poly}(m)\epsilon)$, to a nonzero rational constant times the evaluation subfamily $K^{-1} \psi_{\ell q} \phi_{j-s-1}(b_q)$. Its right inverse therefore has norm at most $\epsilon^{-(s+1)} \text{poly}(m)$. The rational minimum-norm solve, after harmless row rescaling, gives

$$|a_{q,s+1}| \leq \text{poly}(m)/(L\epsilon^{2s+1}).$$

Choose first the fixed exponent in ϵ and then the one in L large enough that $\sum_{l=1}^{s+1} |a_{ql}| \leq 1/2$ for every box. Since $|P_l| \leq 1$ on $[-1, 1]$, all densities have the asserted bounds. Their endpoints and coefficients are rational, and their first s conditional moments are the prescribed polynomial-denominator numbers. Every required global moment is exact. All parameters are polynomial for fixed h, s , as claimed. $\square$

Arithmetic used by a synchronized family of applications. The fixed-minor rounding equations use $KT_{\ell j}$, rather than $T_{\ell j}$ themselves. Thus if several applications use the same rational feature minor, a common grid for the quantities $KT_{\ell j}$ and for all other rounded entries suffices to synchronize their output low-moment denominators. This observation does not itself construct such a shared minor or synchronize affine changes of interval.

Application to cumulative mass and first moment. The relevant fixed feature family is $\psi = (1, r_0, r_1)$, where r_0 is a rational cumulative mass and r_1 a rational cumulative first moment. The ternary Legendre argument conditions this family when its real baseline is

$$1, \quad R = t - \eta H_d, \quad M(t) = \int_0^t u R'(u) du.$$

After sufficiently fine polynomial rational rounding, the empirical Gram hypothesis remains valid. The targets of degree one are fixed rational numbers in the intended uniform-moment construction. The lemma thus produces rational pure boxes with polynomial-denominator masses, endpoints, and means. Their higher moments may have much larger denominators, which is deliberately outside the conclusion.